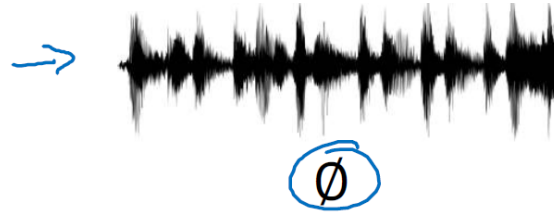


Redes recurrentes

Juan Pajaro

Speech recognition



→ "The quick brown fox jumped over the lazy dog."

Music generation



Sentiment classification

→ "There is nothing to like in this movie."



DNA sequence analysis → AGCCCCTGTGAGGAAGTAG

→ AGCCCCTGTGAGGAAGTAG

Machine translation

→ Voulez-vous chanter avec moi?

→ Do you want to sing with me?

Video activity recognition



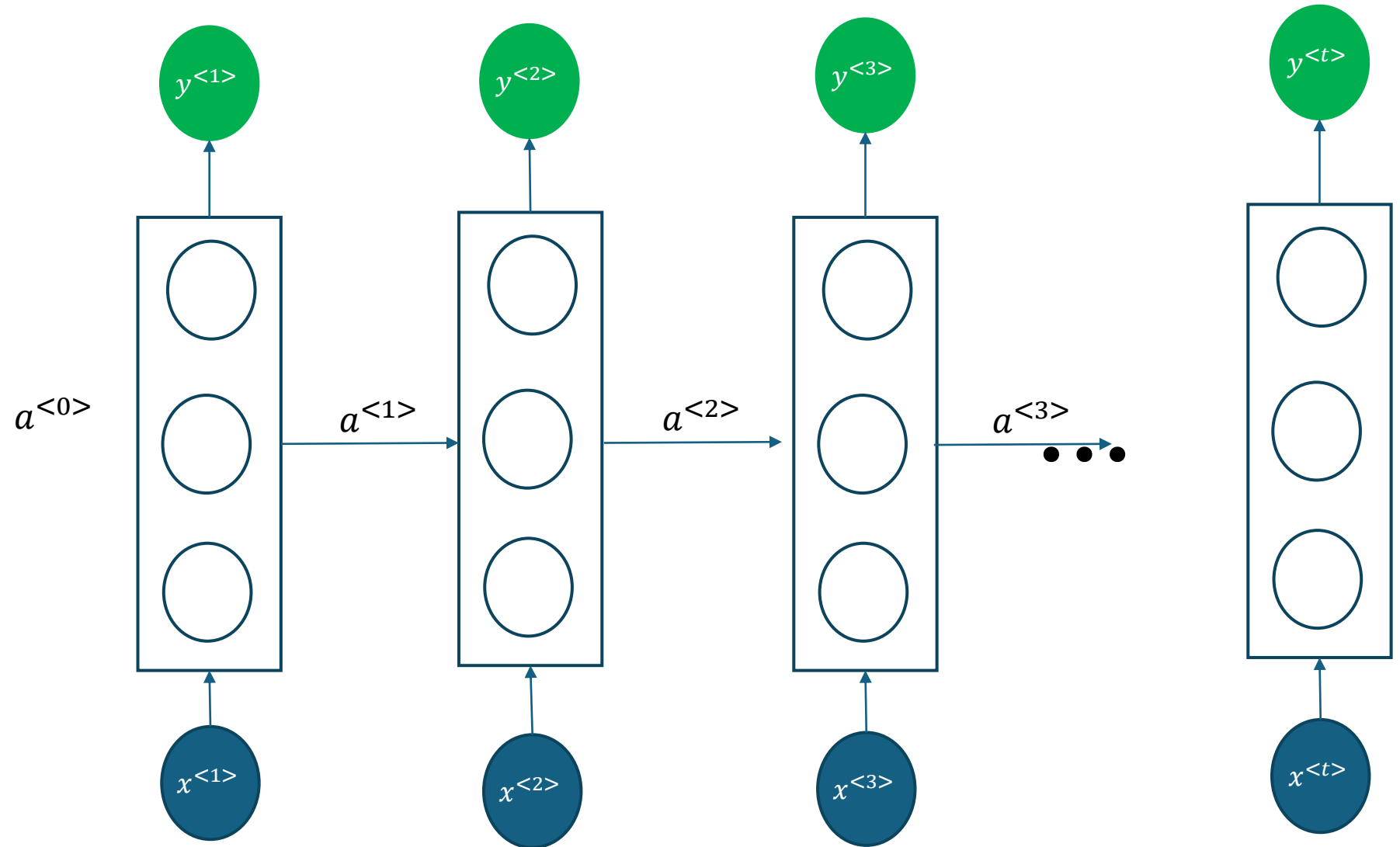
→ Running

Name entity recognition → Yesterday, Harry Potter met Hermione Granger.

→ Yesterday, **Harry Potter** met **Hermione Granger**.

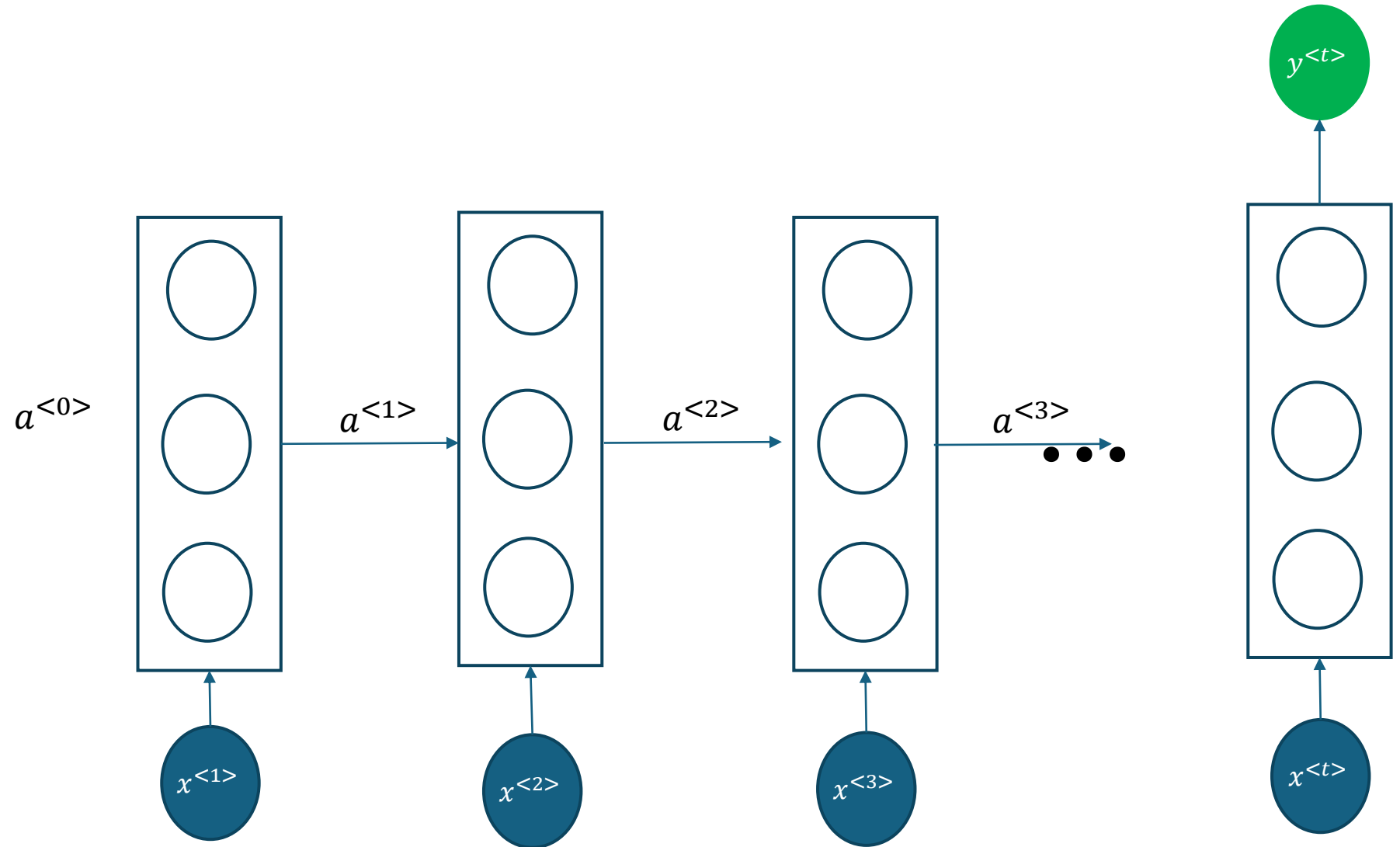
MANY TO MANY

$$T_x = T_y$$



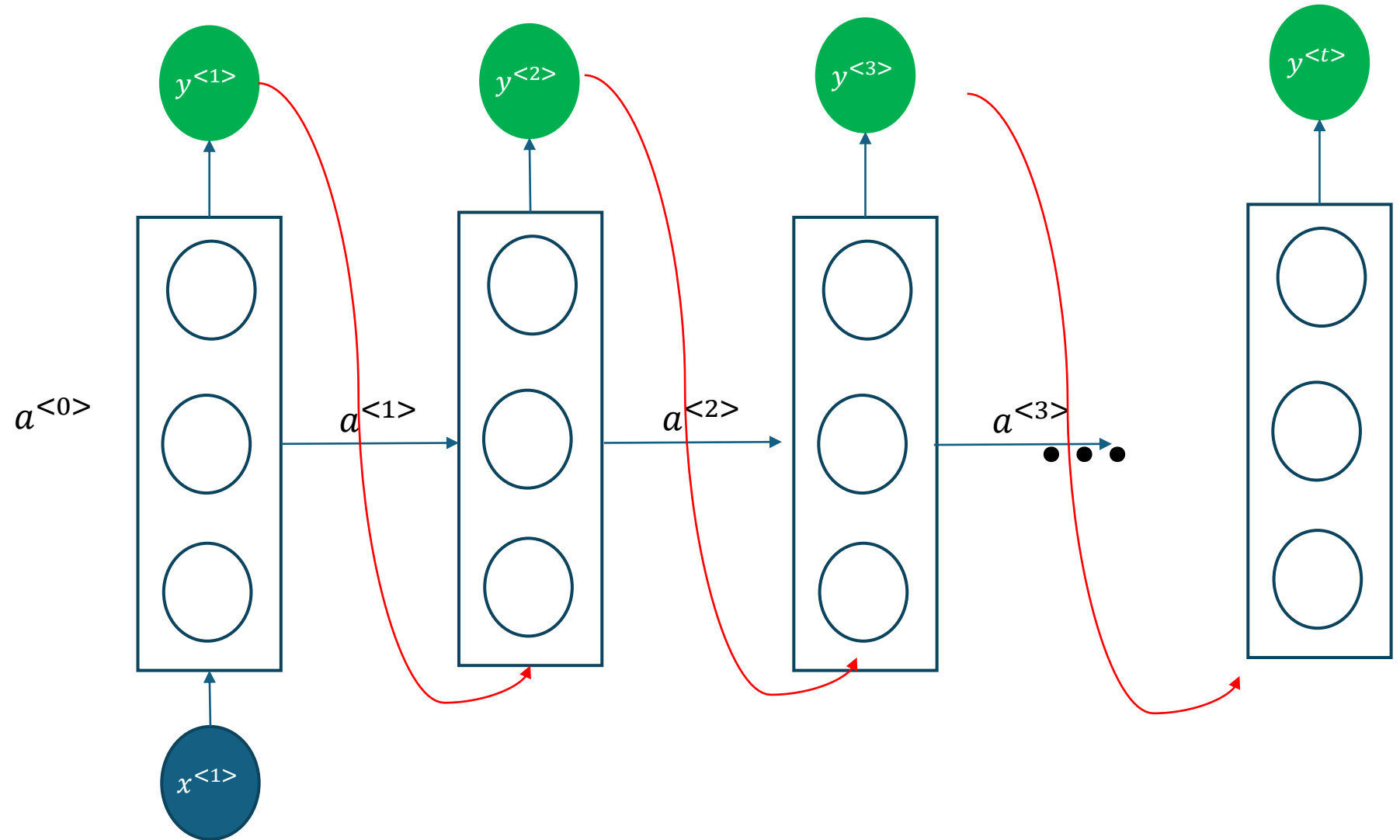
MANY TO ONE

$$T_x \neq T_y$$



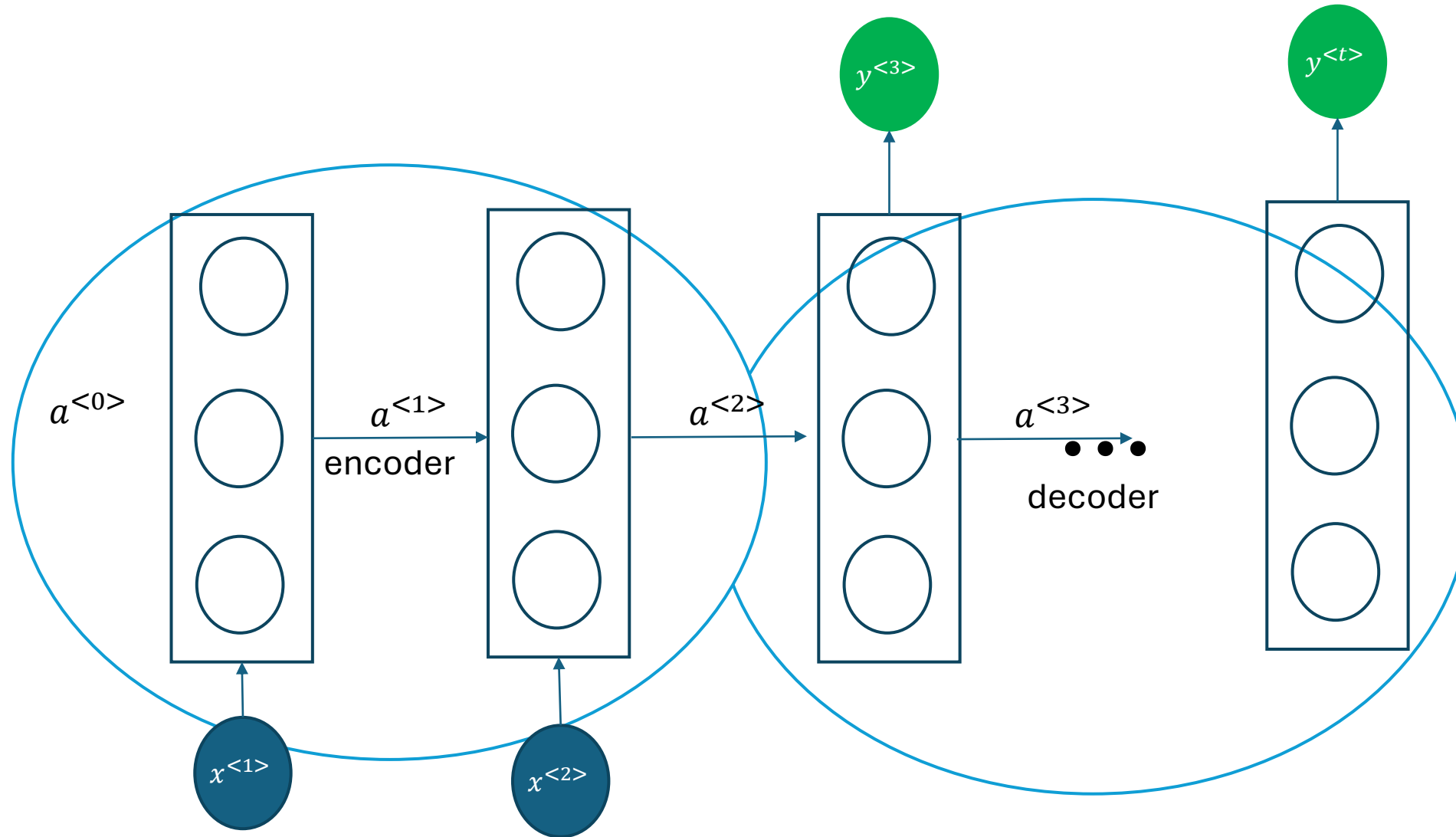
ONE TO MANY

$$T_x \neq T_y$$

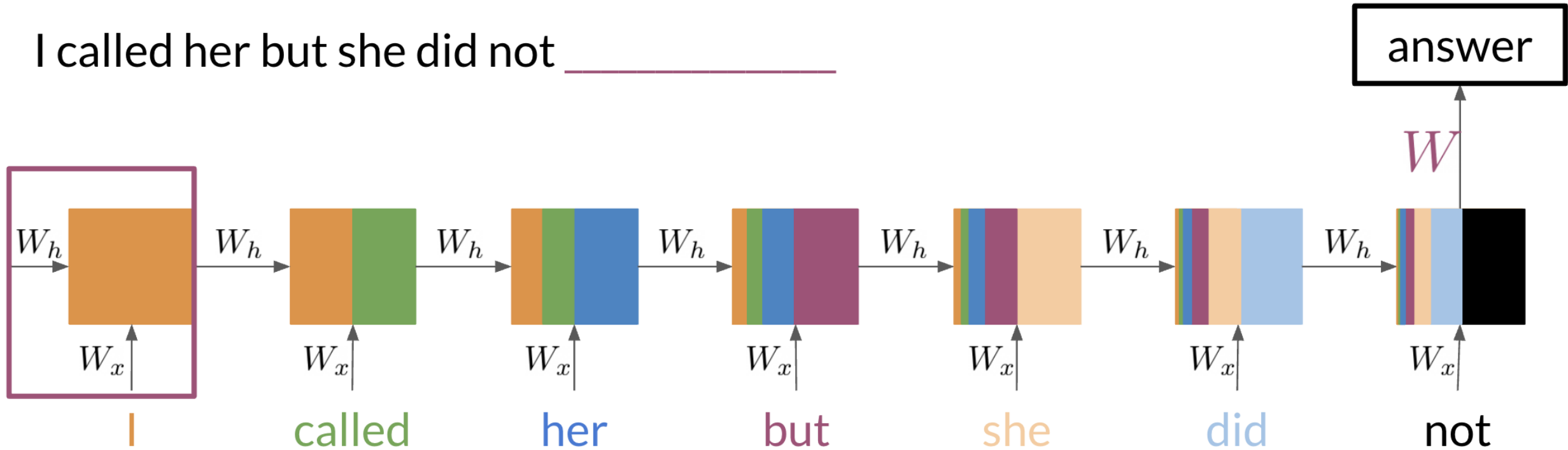


MANY TO MANY

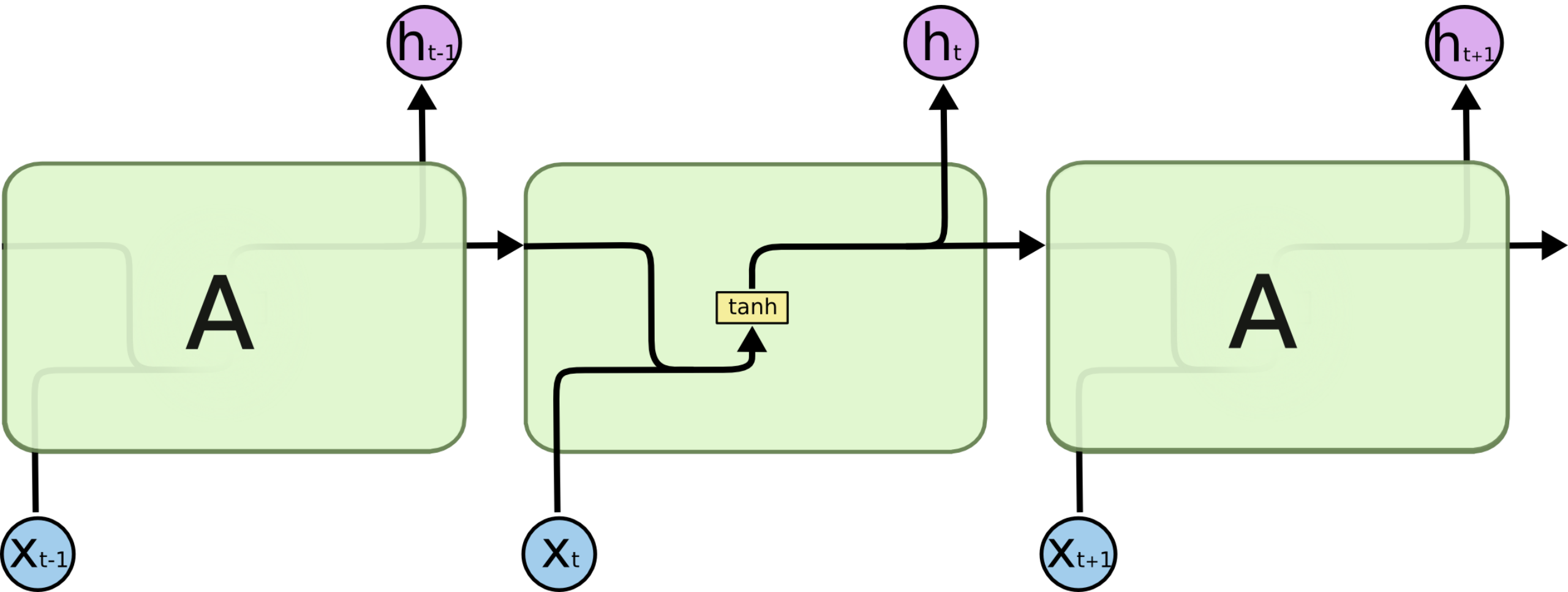
$$T_x \neq T_y$$

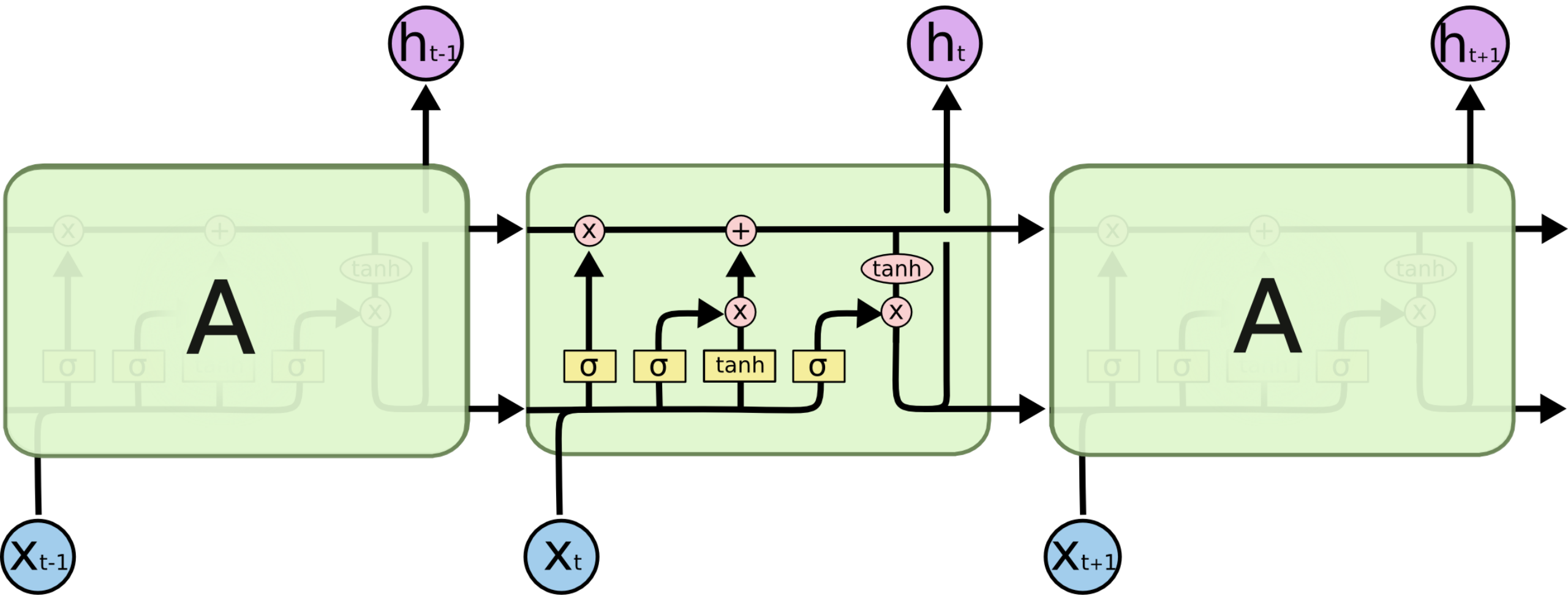


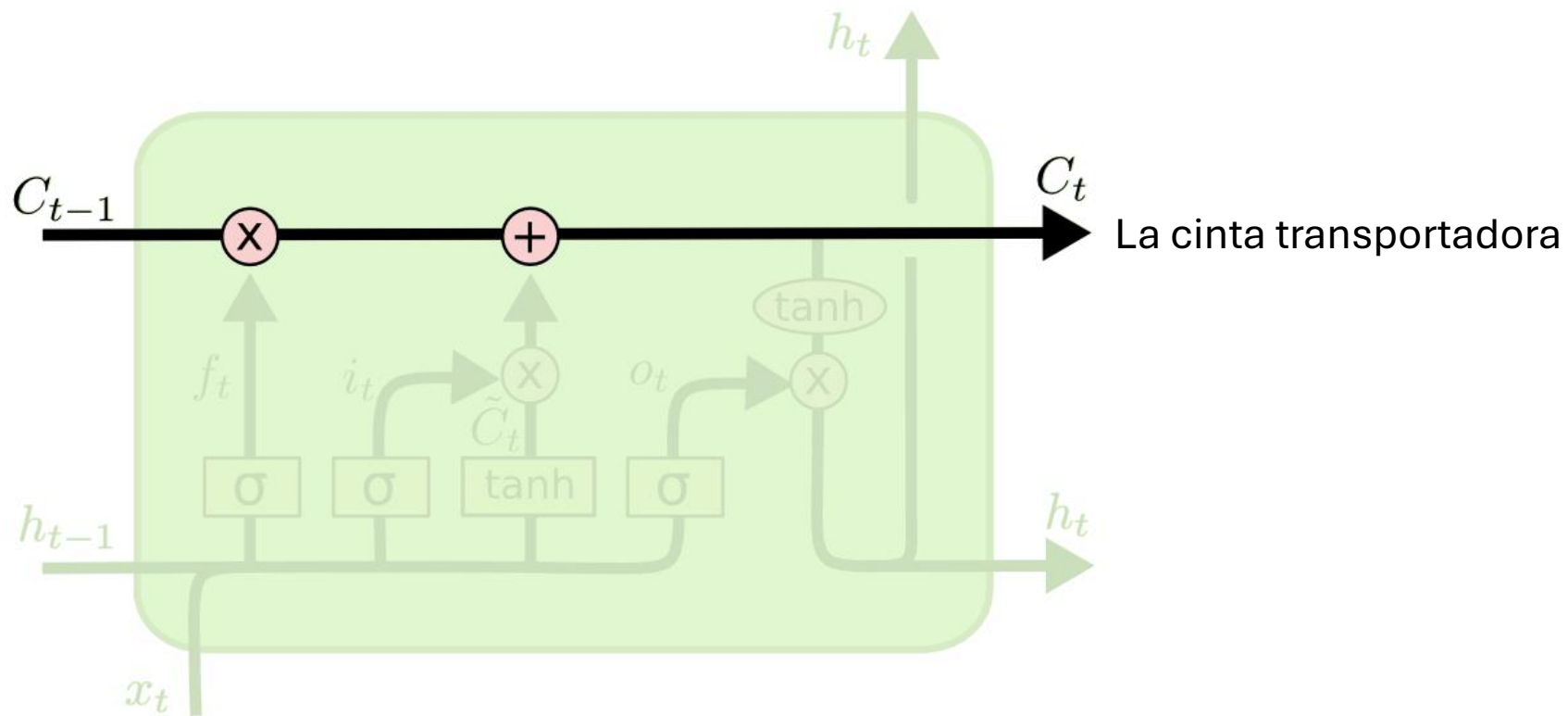
I called her but she did not _____

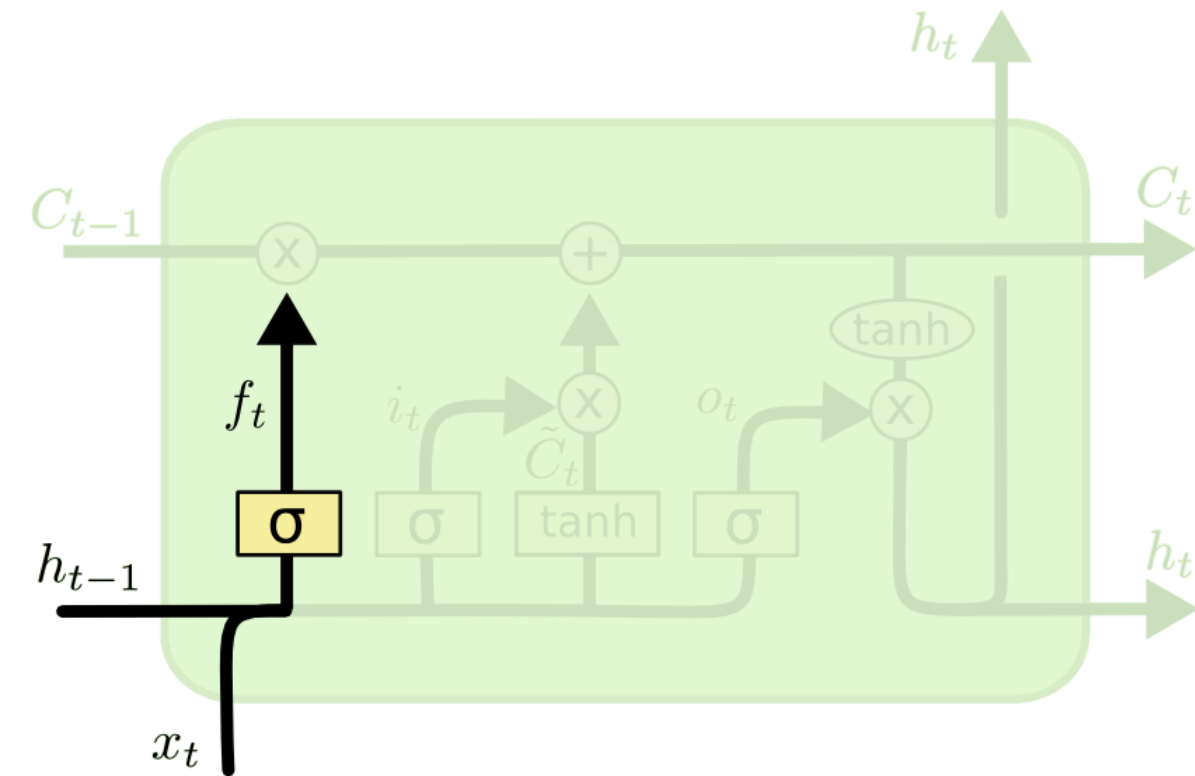


Learnable parameters



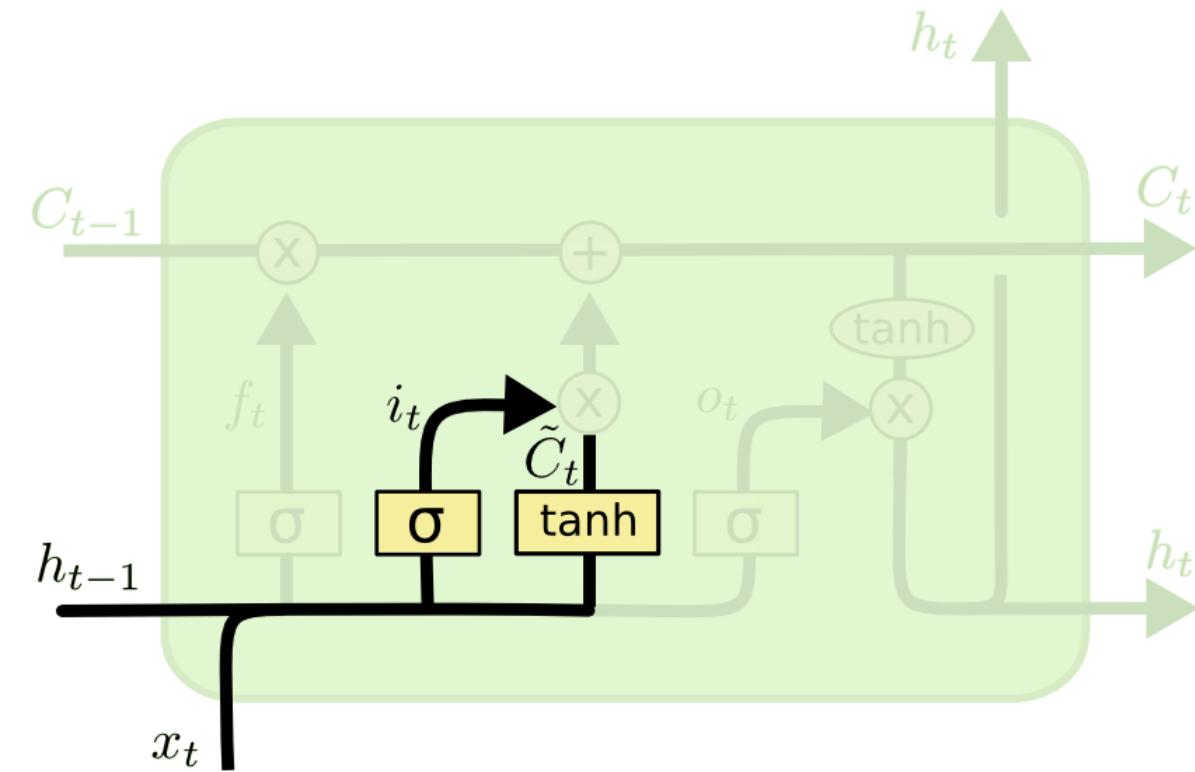






La puerta del olvido

$$f_t = \sigma (W_f \cdot [h_{t-1}, x_t] + b_f)$$

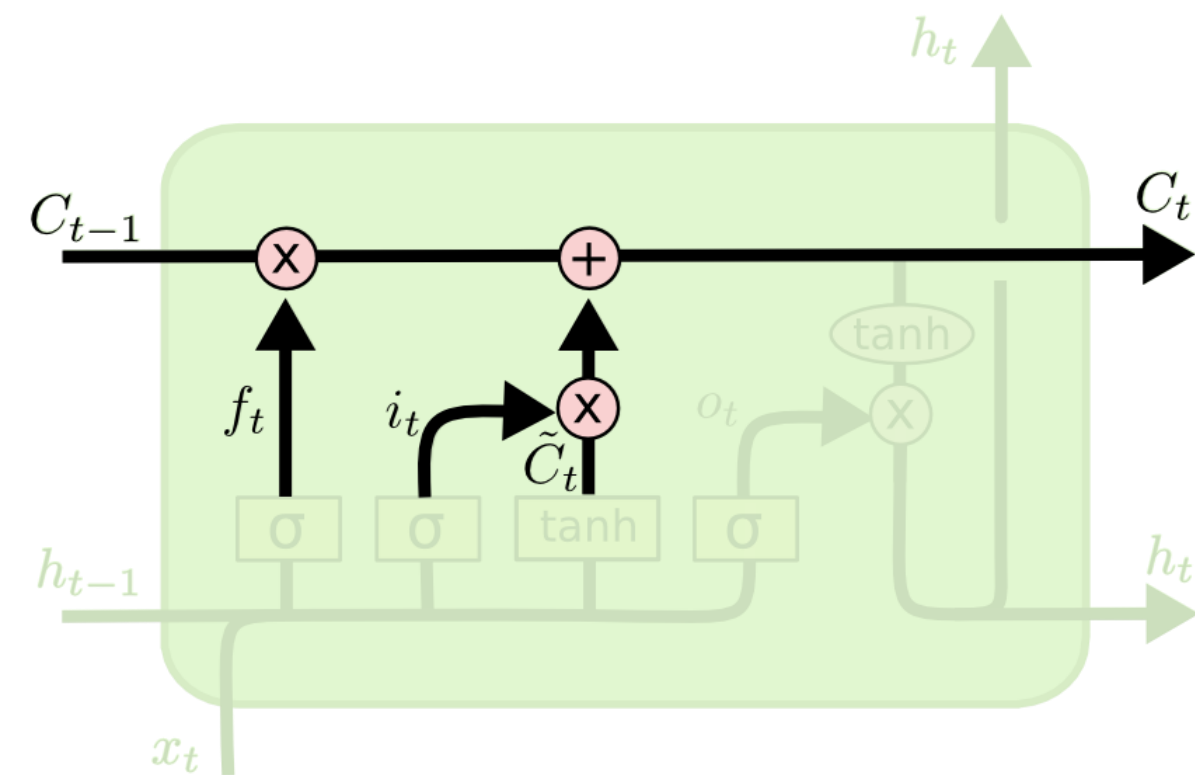


La puerta de entrada

$$i_t = \sigma (W_i \cdot [h_{t-1}, x_t] + b_i)$$

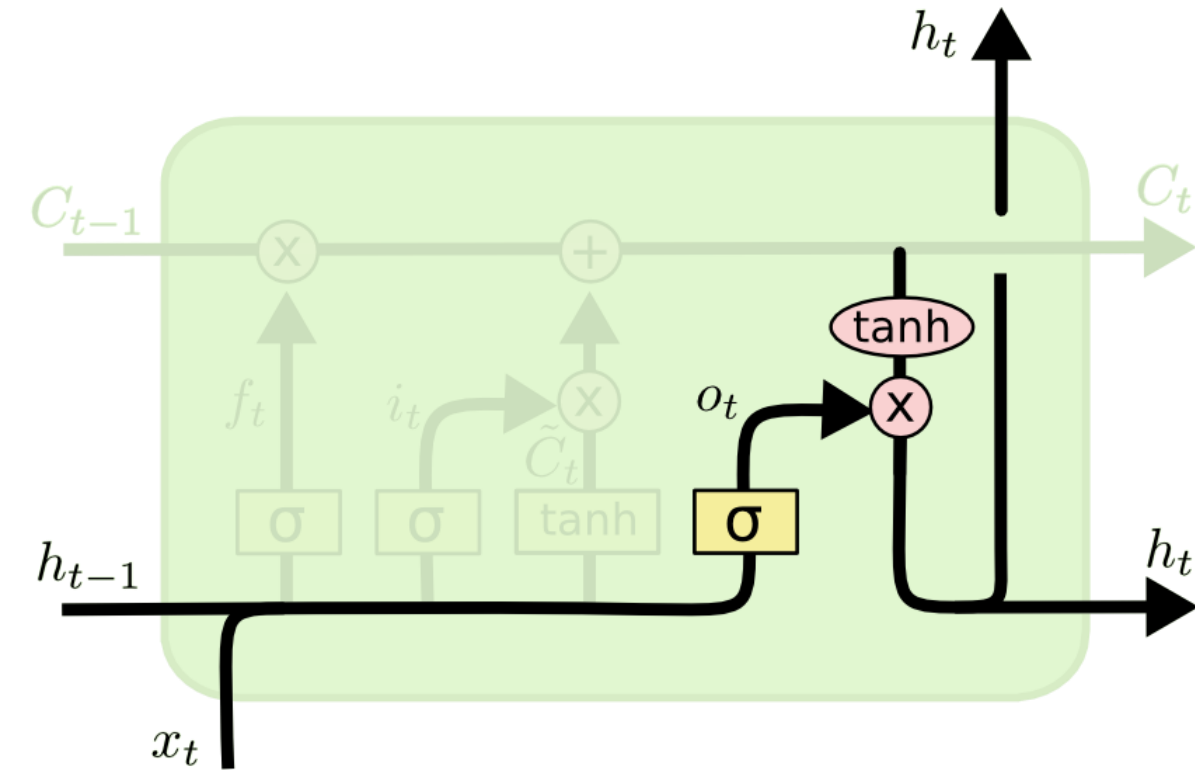
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

La puerta de candidatos



Actualización de la cinta

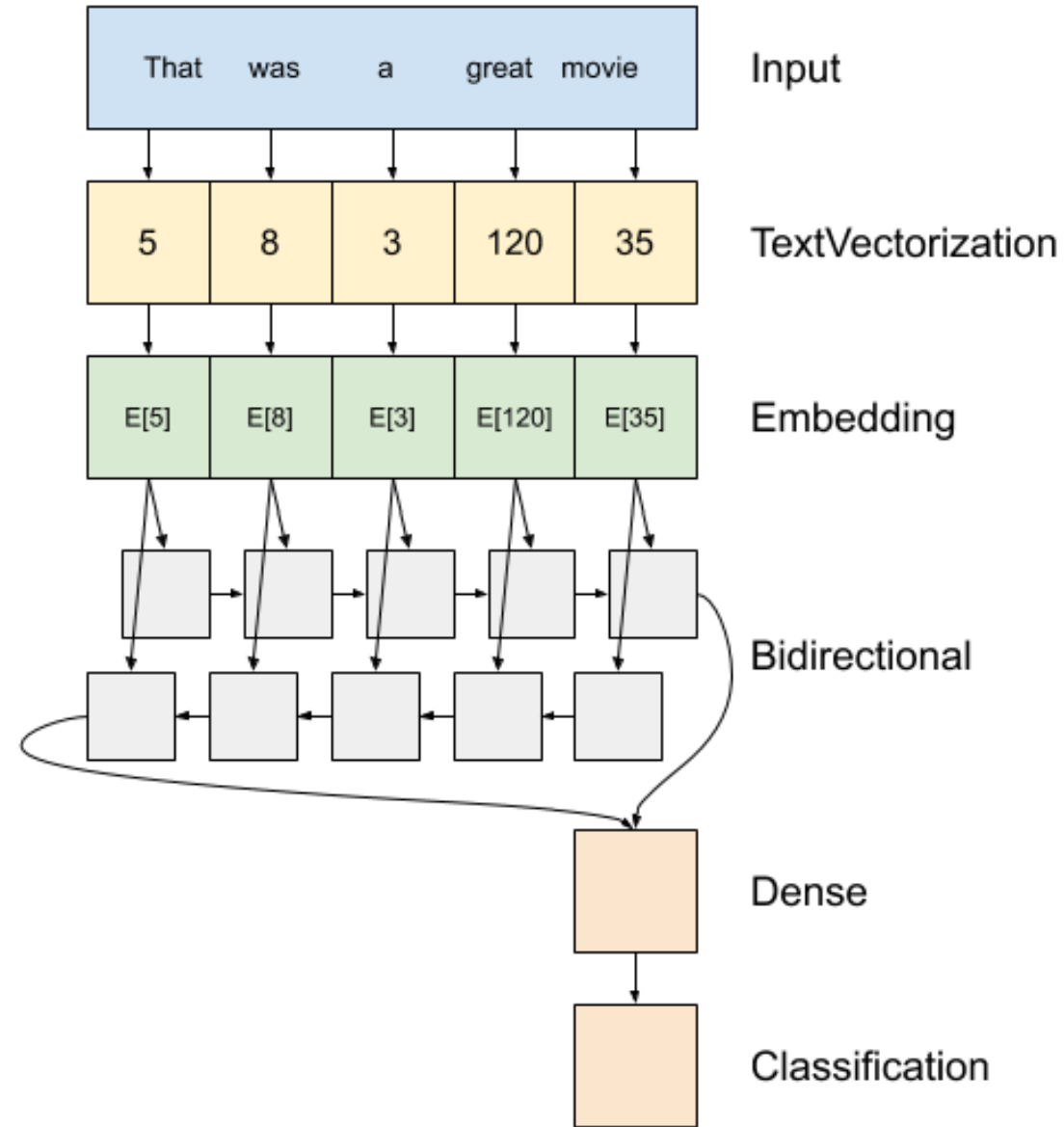
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$



Actualización de los pesos escondidos

$$o_t = \sigma (W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh (C_t)$$





GRACIAS

- <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>
- Speech and Language Processing. (n.d.). Retrieved October 6, 2021, from <https://web.stanford.edu/~jurafsky/slp3/>